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MILITARY SPECIFICATION

TRUCKS, LIFT, FORK, GASOLINE;

GENERAL SPECIFICATIONS FOR

This specification is mandatory for use by all departments and agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers the general requirements for gasoline powered, industrial type fork lift trucks.

1.2 Classification. The classification shall be as specified on the appropriate military specification sheet.

2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on date of invitations for bids form a part of this specification to the extent specified herein. Where there is a conflict with referenced documents, this specification shall govern.

SPECIFICATIONS

Federal

W-B-131 - Batteries, Storage, Lead Acid (Starting Lighting and Ignition - Automotive Type).

Military

MIL-P-116 - Preservation, Method of.

MIL-L-2104 - Lubricating Oil, Internal-Combustion Engine, Heavy-Duty.

FSC-3930

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MIL-F-3541 - Fittings, Lubrication (Hydraulic).

MIL-M-3971 - Meter, Time Totalizing, Operational Use, Electric.

MIL-W-5044 - Walkway, Coating and Matting, Nonslip, Aircraft.

MIL-L-10295 - Lubricating Oil, Internal-Combustion Engine, Sub-Zero.

MIL-B-11040 - Belt, V-Type, Engine Accessory Drive.

MIL-E-11275 - Engine, Gasoline, Industrial Type, 300 BHP and Under, General Specification.

MIL-T-16165 - Interference Shielding, Engine, Electrical System.

MIL-M-21861 - Manual, Technical, Materials Handling Equipment, Preparation, Contents and Approval.

MIL-E-55301 - Electromagnetic Compatibility.

STANDARDS

Federal

FED-STD-595 - Colors.

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-130 - Identification Marking of U. S. Military Property.

MIL-STD-162 - Materials Handling Equipment, Preparation for Shipment, Storage, Cylic Maintenance, Routing Testing and Processing.

MIL-STD-268 - Test and Inspection of Truck, Lift, Fork, Gasoline and Electric.

MIL-STD-826 - Electromagnetic Interference Test Requirements and Test Methods.

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MS-16955 - Hook Type Fork Mounting for Trucks of 2,000 and 4,000 Pounds Capacity Based on 24 Inch Load Center.

MS-16956 - Hook Type Fork Mounting for Trucks of 6,000 Pounds Capacity Based on 24 Inch Load Center.

MS-16957 - Hook Type Fork Mounting for Trucks of 10,000 Pounds Capacity Based on 24 Inch Load Center.

MS-16966 - Tire, Cushion Permissible Sizes and Loading for Use on Original Material Handling Equipment.

MS-16967 - Tire, Solid Permissible Sizes and Loading for Use on Original Material Handling Equipment.

MS-16968 - Tire, Pneumatic Permissible Sizes and Loading for Use on Original Material Handling Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bids shall apply.

Underwriters' Laboratories, Inc.
Publication

Standards for Safety for Internal Combustion Engine Powered Industrial Trucks (UL-558)

(Application for copies should be addressed to: Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, Long Island, New York; 207 East Ohio Street, Chicago, Illinois; or 1655 Scott Boulevard, Santa Clara, California).

American Society of Mechanical
Engineers Publication

American Standard Safety Code for Powered Industrial Trucks (ASA B56.1).

(Application for copies should be addressed to: The American Society of Mechanical Engineers, 29 West 39th Street, New York, New York).

SAE Handbook

(Application for copies should be addressed to: Society of Automotive Engineers, 485 Lexington Avenue, New York, New York).

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National Bureau of Standards
Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402).

3. REQUIREMENTS

3.1 Standard product. Trucks covered by this specification shall be complete and ready for operation when delivered. Unless otherwise specified, batteries shall be furnished with the trucks. Batteries shall be as specified in 3.3.5.7. Trucks shall be constructed and equipped to meet all requirements specified herein. The end items and major and minor assemblies shall be new and of a standard design or model product used by the end item manufacturer current at time of bidding, except for such minor modifications from manufacturer's standard as may be required to conform to these specification requirements. There shall be no change in components to affect interchangeability within individual contract items without authorization by the contracting officer. The component parts of the unit need not be the products of the same manufacturer. However, the manufacturers shall be so established in the industry that prompt and continuing service and delivery of replacement repair parts may be assured.

3.1.1 First article testing and approval. Unless otherwise specified, (see 6.2), before production is commenced, a preproduction model(s) shall be submitted or made ready for examination, testing, and approval, by the contracting officer or his authorized representative. The approved preproduction model may be presented for acceptance as the last unit for delivery under the contract after being rehabilitated to reflect a new appearance. This shall include the furnishing of new tires.

3.1.2 Material. Material not definitely specified shall be at least equal to the end item manufacturer's normal commercial product, and shall meet test requirements specified in section 4. "Normal commercial product" shall be interpreted to mean an item furnished by an end item manufacturer normally engaged in commercial production of the item(s) covered by this specification. The Government shall require any bidder, who proposes to furnish other than a "normal commercial product" or Government specified item to submit, prior to and as a condition precedent to the award of contract, detailed engineering drawings of those parts he proposes to furnish in lieu thereof for approval of the contracting officer.

3.1.2.1 Castings. Castings shall be of the quality normally used in commercial practice, free from blowholes, porosity, shrinkage, cracks, or other defects. Castings shall not be repaired to adversely affect the strength, or intended use. Sufficient finish shall be allowed to permit machining for leak-free sealing at covers, closure points, and fasteners.

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3.1.3 Design. These trucks shall be designed for the rated load, at the specified load center measured from the face of the forks, and at the specified lift height. Design of trucks covered by the specification and all components therein, shall insure safe, efficient, and economical operation under normal service conditions. There shall be no evidence of accelerated wear, or failures, or permanent deformation when operated under normal service conditions.

3.1.3.1 Safety. The truck shall conform to the applicable requirements of Underwriters' Laboratories, Inc., "Standards for Safety" (UL 558, Type G) unless modified herein.

3.1.3.2 Finish. Castings with fastener holes shall provide for proper seating of fasteners such as bolt, lockwasher and nut assemblies, and cap screws.

3.1.3.3 Bearings. Unless otherwise specified herein, rotating parts shall be mounted on ball or roller bearings to the extent normally provided in standard commercial practice. Bearing load capacity, including applicable safety and life factors, as recommended by the bearing manufacturer, shall be equal to or greater than maximum bearing loading under operation required by the test course.

3.1.3.4 Grease fittings. Grease fittings shall comply with Specification MIL-F-3541. Fittings shall be provided where required.

3.1.3.5 Lubrication seals. All lubrication seals shall be properly sized and seated to prevent leakage of lubricant through or around positioned seals.

3.1.3.6 Screw threads. Screw threads shall be in accordance with Handbook H28.

3.1.3.7 Maintainability. Provision shall be made for ready adjustment, servicing, or replacement of fan belt, ignition parts or assemblies, carburetor and components, fuel pump and components, oil filter and components, clutch, starter, generator, generator regulator, battery, wearing parts of steering assembly, tires, wheels, lights, and horn. If hand access openings are used on the exterior of the truck, the edge of the opening shall be smooth and shall be provided with a removable or hinged cover, whenever it is required. There shall be no interference to the servicing or draining of oils and lubricants to or from any assembly or components by frame members or other obstructions, and without requiring the use of flexible connection or other special devices. Drain plugs shall provide for complete drainage. Compliance with specific maintenance requirements for the equipment shall be determined in accordance with Standard MIL-STD-268.

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3.1.4 Certificates of compliance. When specified herein, or upon request, certificate(s) of compliance with the requirements of this specification shall be furnished. Such certificates shall be supported by back-up data which shall be available for review. The certificate of compliance shall read as follows:

> This certifies that the _____(Component, part or assembly) complies with all the provisions and requirements of Specification _____paragraph _____.

The certificate shall be signed by a duly authorized agent to the contractor.

3.2 Propulsion.

3.2.1 Engine. The basic engine less accessories shall be the military type, or when specified (see 6.2), a commercial industrial type, and as described hereinafter. All other requirements including radio suppression, cooling system, fuel system, ignition system and accessories shall be as specified herein.

3.2.1.1 Military. The Military engine shall conform to the liquid cooled group II type of Specification MIL-E-11275 (except thermostat). A model of the engine furnished shall have successfully passed the qualification test requirements of Specification MIL-E-11275. The contractor shall have available a certificate of compliance from the engine supplier indicating that the engines furnished meet the applicable requirements of Specification MIL-E-11275 and that a model of the engine furnished has successfully passed the qualifications listed therein. The certificate of compliance shall list each engine furnished by complete model number and serial number. Unless otherwise specified, production control test, production tests and inspections shall be in accordance with established commercial practice for the type engine to be supplied.

3.2.1.2 Commercial industrial. Unless special power and size requirements are specified (see 6.2), the commercial industrial gasoline engine shall be equivalent in design, power and size to the engine normally furnished by the end item manufacturer in similar commercial products and shall provide sufficient power and torque to meet the performance and operational requirements specified herein. If there is no commercial counterpart, the engine shall be equivalent in design, size and horsepower to engines furnished in competitive commercial models. Engine accessories shall comply with requirements of 3.2.2 engine accessories paragraphs.

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3.2.2 Engine accessories.

3.2.2.1 Choke control. Choke control shall be operated manually. The choke control shall be located on the instrument panel, frame, or body in close proximity and visible to the operator for easy operation.

3.2.2.2 Ignition. The trucks shall have a nominal 12-volt battery ignition system. The system shall include ignition switch, current ammeter, ignition coil, distributor, spark plugs, high tension cables and low tension cables. All wires and cables shall be adequately supported to minimize chafing and provide adequate clearance from moving parts, hot engine parts, exhaust and fuel systems. Cables and wires shall not be supported on oil or grease-retaining surfaces or be exposed to drippage of gasoline, oil or grease. When exposed to unusual vibration, cables, and wiring shall be protected by flexible tubing or suitably harnessed. Grommets shall be provided whenever wiring passes through bulkheads, partitions or structural members, or the wiring shall be suitably fastened to prevent chafing or abrasion. All components of the ignition system shall be weather resistant to prevent the entry of moisture when operated or stored outdoors under all weather conditions. The ignition system shall not be protected by fuses.

3.2.2.2.1 Distributor. The distributor shall be equipped with a fully automatic centrifugal, vacuum or combination centrifugal vacuum control to advance or retard the spark.

3.2.2.2.2 Ignition coil. The ignition coil shall be mounted in a position isolated from excessive engine heat sources, and shall be easily replaceable without removal of any major components.

3.2.2.3 Generator or Alternator. The battery-charging generator shall be of the 2-brush, open type with not less than 18 ampere rating, or an alternator with at least equal charging characteristics, drive by "V" belt(s) conforming to Specification MIL-B-11040. Generator output shall start at manufacturer's recommended idle speed. The generator shall be cooled by air drawn through openings in the generator by a fan mounted on the armature shaft. When an alternator is provided positive markings shall be provided at the battery terminals to prevent applying reverse polarity to the electrical system.

3.2.2.3.1 Regulator. The generator regulator shall be a three unit type, consisting of a cut-out relay and voltage regulator and a current regulator. When an alternator is used the combination of an alternator and regulator shall provide equal protection to the electrical system as the generator regulator. The regulator shall be conveniently located for ease of adjustment and maintenance. The cut-out relay shall open

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when the generator voltage falls below that of the battery. The voltage regulator unit shall control the generator voltage within a range of 13.8 to 14.8 volts by generator field control. The current regulator unit shall limit current output to that of the generator rating. A screw or other adjusting device to regulate the charging voltage shall be provided. The adjusting device may be accessible either before or after seal is broken.

3.2.2.4 Starting system. The starting system shall include a starting motor solenoid, a starting motor, necessary wiring and cable and starting motor switch. Trucks equipped with power shifted transmissions shall be provided with an interlock in the starting system or other means to prevent energizing the starter motor except when the directional control lever is in the neutral position. Cranking torque of the starting motor shall be sufficient to start the engine with battery specified for the truck when both engine and battery are thoroughly cold soaked, at 0°F. A certificate of compliance is required.

3.2.2.5 Fuel system. The capacity of the fuel tank shall insure not less than 4 hours continuous operation as measured by Standard MIL-STD-268. The tank shall be equipped with a safety filler cap assembly painted with red enamel which shall incorporate a self-closing cap, screw-locking device for cap, removable strainer and provide for padlocking. The cap assembly and the neck of the fuel tank shall be threaded and when installed, the cap shall not be removable without the use of tools. A standpipe or other suitable means at the fuel line inlet shall be provided to prevent foreign matter from entering the fuel line and to prevent drawing fuel from bottom of tank. The fuel tank shall be equipped with a drain plug at the lowest point of the tank. The drain plug shall be removable with hand tools without the removal of any other component. The fuel tank shall be protected by position or otherwise from external damage. A shut-off valve shall be accessibly located immediately adjacent to the fuel tank. Adequate fuel lines shall be provided to insure continuous operation at full throttle and maximum loads, including slope ascension. Fuel lines shall be protected to the extent necessary for preventing vapor lock by suitable insulation or heat reflectors when located near high temperature areas of the engine. Other construction and design details shall be in accordance with Underwriters' Laboratories, Inc., "Standards for Safety" (UL 558, Type G). No provision for auxiliary fuel supply is required. Fuel pump shall be of the automotive diaphragm type. A fuel filter and a metal sediment bowl of adequate capacity shall be provided between the fuel supply and the pump and shall be accessible from above or from the side. The bowl shall not be obstructed and shall be located to minimize fire hazard. Clearance shall be provided for removing and cleaning the filter.

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3.2.2.5.1 Carburetor. The carburetor shall be of the aspirating or injection type. Provision shall be made for external adjustment of the idle metering system and idle limit stop. The main metering system shall be controlled by removable fixed orifice jets designed for sea level operation.

3.2.2.5.2 Air cleaner. The carburetor air inlet shall be provided with an air cleaner. Oil bath air cleaners shall be of the automotive type having a capacity sufficient to allow the engine to operate under all conditions specified herein and in Specification MIL-E-11275, without drawing oil from the air cleaner at wide open throttle. The air cleaner shall be located or designed to prevent water entering the cleaner when the vehicle is operated. The air cleaner shall be constructed so as to permit disassembly without tools for servicing. It shall be accessible when engine cowl, hood, or access plate is opened. When dry type replaceable cartridge type air cleaners are permitted, they shall be of the automotive type.

3.2.2.6 Cooling system. The cooling system shall be of the closed pressure type, incorporating a radiator with a radiator pressure cap, thermostat, fan, and circulating pump. The cooling system shall have sufficient capacity to prevent coolant temperature of 115°F. over any ambient temperatures from 20°F. to 104°F. when the truck is operated continuously with the rated load. One drain cock shall be located in the base of the radiator or pump inlet elbow, whichever is lower, and the second shall be located at the lowest point of the liquid coolant jacket in the cylinder block.

3.2.2.6.1 Fan. The fan shall discharge away from the operator and rotate on antifriction bearings and be driven by a "V" belt(s) conforming to Specification MIL-B-11040. Fan belt(s) take-up shall be at least three (3) percent of the belt circumferential length, plus an allowance for replacement of a new belt(s) without forcing. The belt(s) shall be clear of all other components within the adjustment range.

3.2.2.6.2 Water pump. The water pump(s) shall be of the centrifugal type, shall be replaceable, and shall be equipped with shaft seals that will prevent leakage. The shaft shall be corrosion resistant when exposed to water and a mixture of ethylene-glycol and water. A lifetime lubrication system shall be required or an adequate and readily accessible pressure lubrication fitting shall be provided.

3.2.2.6.3 Thermostat. A thermostat of nominal temperature between 165°F. to 180°F. shall be located in the cylinder head outlet to the radiator. It shall be easily removable. Thermostat located in the radiator hose is not acceptable.

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3.2.2.6.4 Radiator. The radiator assembly shall consist of a fin and tube type core with an expansion tank and sediment tank. Where the top of the radiator is lower than the top outlet of the engine, a surge tank may be used. The radiator cap shall be accessible without physical removal of engine hood. The radiator shall be mounted to prevent its being damaged due to shock and racking experienced in normal operation of the equipment. The radiator shall be protected by a heavy grilled guard, or the counterweight. A shield shall be provided, if necessary, to protect the radiator from mud and gravel thrown by the steering tires. Hoses shall have adequate clearance of the truck structure or other components under all operating conditions.

3.2.2.6.5 Gaskets, seals and packing. Gaskets, seals and packing used in the cooling system shall be made of materials capable of resisting deterioration from mixtures of water and ethylene-glycol-base antifreeze.

3.2.2.7 Governor. The governor shall be adjusted to limit the maximum speed of the engine to not in excess of the engine manufacturers' published recommended maximum engine speed for industrial trucks application. The governed speed of the engine shall limit the maximum speed of the vehicle as required in 3.6.5.

3.2.2.8 Exhaust system. The exhaust system shall conform to the structural and safety requirements of Underwriters' Laboratories, Inc., "Standards for Safety Power-Operated Industrial Trucks (UL 558, Type G, as applicable)". The exhaust system shall be protected against entry of rain and shall include drainage provisions to prevent accumulation of water and condensed vapors. The exhaust system shall terminate near, but not beyond the plan outline of the truck. Its outlet shall be no more than 40 inches above ground level. The muffler shall be independently supported around the body or muffler outlet. If the tail pipe is independent of the muffler or extends more than 12 inches from the muffler, it shall be supported. Exhaust gases shall be emitted in a horizontal direction at the rear or side rear of the truck or shall be dissipated into the fan discharge away from the operator. The exhaust system shall be arranged to prevent burns to the operator during normal operations. The discharge shall be so designed that a shop exhaust system may be conveniently connected and used when the engine is operated.

3.2.2.8.1 Spark arresting device. The exhaust system shall meet (UL 558, Type GS) requirements for dry type mufflers when specified (see 6.2).

3.2.2.9 Crankcase emission system. Crankcase emission shall be directed into the intake manifold at the riser. If air is drawn into the crankcase, it shall pass through an air filter prior to entering the crankcase. Provision shall be made to regulate the air flow under all engine operating

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conditions by means of a metering valve. Regulation shall be adequate to ensure that, within normal carburetor adjustment limit, the operation of the engine is not affected by the ventilation flow. The complete system shall be arranged to fit within the existing engine enclosure with no restrictions or traps in the air lines and shall be accessible for maintenance. Air between the crankcase and the metering valve may be either filtered or unfiltered.

3.2.2.10 Fungus and moisture proofing. When specified the equipment shall be treated to resist fungus and moisture. All electrical equipment and wiring including that on the engine shall be treated in accordance with the fungus and moisture resistance requirements of MIL-E-11275.

3.2.2.11 Oil filter. Oil filters of either the by-pass or full flow type shall be used. The oil filter shall be mounted in an accessible location for ease of replacement of elements. Oil lines (if applicable) to and from the filters shall be installed to minimize fatigue from vibration.

3.2.3 Drive assembly. The drive assembly shall consist of all components necessary to transmit power from the engine to the wheels. Each gear, shaft, and axle shall be of heat treated alloy steel. All rotating shafts and axles shall be supported on anti-friction bearings. All gears shall operate in lubricant, and a standard pressure grease lubrication system shall be provided for all friction parts not so lubricated. Gear housing shall be equipped with filler plugs and magnetic drain plugs. Differential ring and pinion gears, when used, shall be adjustable. All levers to gear enclosures shall be provided with seals, caps, rubber boots or similar means to prevent the entry of water into the enclosure under normal operation or outside storage. When a drive shaft is provided there shall be a slip joint integrated into the drive line to facilitate the removal of the drive shaft. With drive wheels locked, the drive assembly shall be capable of withstanding a 150 percent of maximum bare engine torque as published by engine manufacturer, multiplied by the stall torque rate of the torque converter applied to the input shaft to the transmission without failure of any component. The hydraulic pressure may be increased or the clutch locked by external means to prevent slippage at the discretion of the truck manufacturer. In all cases, the transmission shall be in lowest gear range. When test is required (see 6.2), the drive assembly may be tested before or after truck is completely assembled.

3.2.3.1 Transmission. Unless otherwise specified a power shifted transmission shall be furnished (see 6.2).

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3.2.3.1.1 Power shifted transmission. The power shifted transmission shall be of the continuous drive type within each speed range(s). Trucks having a capacity of 10,000 Lbs. and above shall have not less than two speed ranges forward and reverse. An oil filter is required for the transmission oil. Means shall be provided to check the transmission oil level Transmissions shall provide for positive inching control of truck (throughout the entire range of engine speed) in the forward and reverse directions. The inching control shall permit tilting and full lifting speeds with rated load independently of and simultaneously with truck motion while in inching travel. The inching control shall be capable of meeting the requirements outlined in Standard MIL-STD-268 without application of the brakes.

3.2.3.1.2 Mechanical transmission. When a mechanical transmission is specified it shall be one utilizing a hand operated lever for gear shifting between speed ranges and foot operated mechanical clutch to disengage the engine from the transmission. The clutch shall be of the proper design and characteristics to withstand stresses induced by the engine and transmission and shall provide smooth acceptance of power when clutch is operated. Clutch shall be disengaged by depressing a hobbed or rubber capped pedal. Force required to depress the clutch pedal and disengage clutch shall not exceed 40 pounds. An easily accessible lubrication fitting for the clutch throwout bearing and clutch shaft shall be provided, so that the bearing may be lubricated without removal of floor plates or other related components. If self-lubricating bearing is used a plug shall be used in place of the fitting to provide for lubrication of non-self lubricating replacement bearings.

3.2.3.2 Torque converter. A torque converter shall be supplied with the power shifted transmission.

3.2.4 Hydraulic system. Power used for hydraulic functions shall be furnished by a hydraulic pump(s) driven by the engine or transmission. The lifting system shall provide for lowering of the load at a rate of not more than 80 feet per minute in event of failure of or damage to hydraulic hose supplying the lift cylinder. Hydraulic reservoir and system shall be capable of being drained without spillage. Hydraulic components shall provide protection against contamination of hydraulic fluid by direct impingement or seepage of other fluids. All cylinders shall be provided with "wiper rings" to preclude entry of dust and dirt into the packing. The reservoir shall be of welded construction, free of metal particles or other foreign material, and have a capacity of not less than 110 percent of the fluid required for the hydraulic system operation . Fluid in the hydraulic system shall conform to MIL-L-2104, grade 10, or MIL-L-10295. The reservoir shall have sufficient capacity to prevent air entering the system with all hydraulic pistons fully extended and sufficient free air capacity to prevent oil being discharged through the reservoir air vent when maximum return flow of oil is surged

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into the reservoir from the system. All piston rods shall be provided with a hard chrome plate finish. Surfaces shall be thoroughly cleaned prior to plating in accordance with normal commercial practice. Plating shall be electro-deposited hard chromium finish, having uniform thickness of not less than 0.0005 of an inch. A removable strainer of 50 mesh opening shall be installed in the suction line between the hydraulic reservoir and the pump. A removable 25 micron full flow or by-pass filter shall be provided to filter all the oil and may be installed in either the suction line between the reservoir and the pump or in the return line between the valve and the reservoir. If the filter is of the full-flow type, it shall have an emergency by-pass. When the micron filter is installed in the suction line its relative position with respect to the 50 mesh screen is optional but it shall not by-pass the 50 mesh screen. A 1/4 inch pipe tap shall be provided in either the pressure inlet side of each valve or in a tee between the pump and the valve to check pressures in the hydraulic system. Hydraulic hoses shall be properly restrained to protect them from unnecessary flexing, pressure surges, or contact with other moving mechanical components. Means shall be provided to indicate the required level of the hydraulic oil.

3.2.4.1 Hydraulic fitting, hoses, and tubing. All components utilizing hydraulic fluid from the hydraulic reservoir excluding brakes and steering systems shall conform to the following standards. Hydraulic fittings used on valves and cylinders shall be the "O" ring straight thread type in accordance with SAE J514 but not limited to the fittings listed therein. When hydraulic tubing is used, fittings shall conform to SAE J514 37 degrees two or three piece tube assembly. All other fittings shall be of the 4 bolt split flange type in accordance with SAE J518 or the "O" ring straight thread type except for barbed or clamp types at both ends of suction, return, and cylinder bleeder lines, and dry seal connectors on lines where hydraulic pressure does not exceed 200 psi. Hose fittings shall be of the reusable screw-on type 37 degree flared type female in accordance with SAE J516, except for the hydraulic hose flanged "O" ring connections which shall conform to SAE J519. Hydraulic hose sizes shall be restricted to the following sizes: 1/4", 5/16", 3/8", 13/32", 1/2", 5/8", 3/4", 1", 1-1/4", 1-1/2", 2", and 2-1/2". Hydraulic hose specification with respect to applications shall conform to the following:

SIZE	APPLICATION	HOSE SPECIFICATION
1/4" - 3/4"	Suction and discharge	Double cotton braid SAE 100R3
1" - 2-1/2"	Discharge	Double cotton braid SAE 100R3 or equivalent
3/4" - 2-1/2"	Suction	SAE 100R4
1/4" - 1/2"	Pressure	SAE 100R1 or SAE 100R5
1/2" - 1"	Pressure	SAE 100R2

3.3 Structure.

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3.3.1 Chassis and frame. The frame and related structure shall be fabricated from steel and steel castings at least equal to the size and design of manufacturer's normal commercial frame members and shall be capable of withstanding 300 percent of rated load without permanent deformation. Frame members, bracings, and all their joints shall provide a rigid unit structure. The frame and related structure shall have sufficient depth to afford protection for working parts and shall be capable of withstanding, without permanent deformation, stresses induced by capacity loads and tests in Standard MIL-STD-268. Unless otherwise specified (see 6.2), the complete frame structure shall be tested as outlined in Standard MIL-STD-268.

3.3.1.1 Uprights and carriage assembly. Uprights shall be of the telescopic type with sufficient strength and rigidity to withstand, without permanent deformation, stresses induced by capacity loads and tests in Standard MIL-STD-268. Uprights shall be positioned in such a manner that the back surface of lower fork mounting plate or cross bars in the vicinity of the tires is located forward of the vertical plane tangent with the front face of the drive tires. The lift cylinder shall be mounted and positioned so that it does not protrude forward of the plan outline of the uprights. Chains, when used, shall be installed in such a manner that they can not be displaced from the roller until they have been raised a minimum of 3/8". Cross members of the mast structure, when in a lowered position, shall not interfere with the operator's horizontal or downward line of sight directly to the front of the vehicle. The operator shall be able to see the tip of at least one fork at any lift height without leaving his seat when there is no load on the forks. The lift shall be hydraulically actuated, with the upper portion automatically lifting as the load is raised above the "free lift" position. Adequate means shall be provided in the uprights to compensate by replacement or adjustment for wear which may develop between sliding surfaces or lateral rolling contacts. Positive means shall be provided to prevent overtravel of the carriage or channel in both upper and lower position. Carriages for trucks equipped with side-shifters shall be supplied with adjustable thrust bearings to resist thrust introduced by off-center loading. Chains used in the lifting mechanism shall be a listed ASA Standard.

3.3.1.2 Forks. Forks shall be capable of withstanding, without permanent deformation 300 percent of rated load, as well as stresses induced by carrying out all applicable provisions of Standard MIL-STD-268. Forks shall be in accordance with the military standard sheets and the dimensions as specified on the applicable specification sheet. Forks shall be attached to the fork carriage in a manner which prevents lateral slippage or permanent deformation of fork carriage components, when subject to the applicable provisions of Standard MIL-STD-268. Unless otherwise specified, forks on trucks up to but not including 10,000

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pounds capacity shall be adjustable laterally without the use of hand tools. Trucks 10,000 capacity and above shall provide for adjustment of the forks simultaneously by auxiliary means, i.e. crank and screw or similar device, from one side of the fork carriage by one man without undue effort and the adjusting mechanism shall not bind or break under normal usage. Undue effort shall be classified as an effort of more than 20 pounds at the handle. Protruding handles, levers, etc., shall fold downward or inward in order to eliminate interference with truck operation. The adjusting worm shall be protected to prevent foreign matter, such as dirt and mud collecting on its surfaces.

Fork thickness shall not exceed 1/2 inch at the tips and the end taper shall be not less than 14 inches long. Taper shall be on the underside of the forks. The dimensions and spacing adjustment shall be measured between outer edges of the forks.

3.3.1.3 Load backrest. A metal backrest, removable without disturbing any other component, shall be provided. Load backrest shall be flush with the forward vertical surface of the forks. In conjunction with fork and hanger design it shall provide a vertical rear guard at least 48 inches high measured from the load carrying surface of the forks and at least equal to the width of the carriage. It shall have strength and stiffness sufficient to prevent a rated load 48 inches high or less from forcing backrest against the uprights when the mast is at maximum rearward tilt. The backrest shall be firmly attached to the lift carriage. It shall not unduly obstruct operator's visibility. The spacing of vertical members shall be no more than 8 inches between members. The backrest shall be capable of withstanding, without permanent deformation, a horizontal static test load equal to the rated load uniformly distributed over the area from the load carrying surface of the forks to the top of the backrest. The removable backrest may be eliminated if fork and hanger assembly will provide protection equal to the above.

3.3.2 Wheels and tires.

3.3.2.1 Wheels for solid tires. The truck shall be mounted on steel, malleable iron, or cast iron wheels, equipped with tapered roller bearings. Wheels, when equipped with new tires, shall clear any part of the truck structure and mast structure by at least 1/4 inch under all conditions of operation over the test course described in Standard MIL-STD-268.

3.3.2.1.1 Solid tires. The tires shall be new and unused and shall be selected from sizes listed on MS-16966 or MS-16967. The tires shall be of the industrial, solid rubber, or cushion type, molded onto a new press-on type steel band. Front tires shall have commercial, non-directional type tread; rear tires may have either smooth or non-directional type tread. Tire loadings shall not exceed the values shown on MS-16966 or MS-16967 when tested in accordance with Standard MIL-STD-268.

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3.3.2.2 Wheels for pneumatic tires. The truck shall be mounted on steel, malleable iron, or cast iron wheels, equipped with tapered roller bearings. Wheels shall have demountable type rims which can be removed from hubs without deflating tires. Steer wheels, when equipped with new tires, shall clear any part of the truck structure by at least 3/8 inch. Drive wheel clearance to any part of the truck structure shall be sufficient to permit the use of standard automotive tire chains. Spacings between rims of dual tired trucks shall be such as to prevent contact of the tires when truck is on level surface with rated load.

3.3.2.2.1 Pneumatic tires. The tires shall be new and unused and shall be selected from sizes listed on MS-16968. The tires shall be pneumatic, high pressure, tube type, furnished with non-directional tread type. Tire loadings shall not exceed the values shown on MS-16968 when tested in accordance with MIL-STD-268.

3.3.2.1.1 Tubes. Tubes shall be heavy duty type and the size required to fit the tires furnished. Valve stems shall be arranged to permit inflation of mounted tires from the side of the truck. When split rims are furnished, a canvas or rubberized flap shall be provided between the tube and the rim for protection of the tube.

3.3.3 Steering. An automotive type steering wheel not to exceed 20 inches in diameter, mounted on the steering column, shall turn the steer wheels. Wheels shall steer true in accordance with the Ackermann steering principle. Clockwise rotation of the steering wheel shall provide for right-hand turning of the truck while in forward motion. Steering force shall be in accordance with 3.6.4. When power steering is required a hydraulic steering booster or full power steer shall be supplied. Provision shall be made for precision adjustment for wear and alignment of all major steering components and for lubrication of all friction points by accessible grease fittings or a lubricant reservoir. When provided, the reservoir shall be accessible for filling and checking of lubricant level. The truck shall negotiate the obstacle course in Standard MIL-STD-268, with only the displaced wheel off the ground at any time. Means shall be provided to prevent wheels or tires or steering linkage from contacting any part of truck structure, other than stops, irrespective of obstacle magnitude or cramped steer angle. Spindles and kingpins, if used, shall be of heat-treated alloy steel.

3.3.4 Brakes.

3.3.4.1 Service brakes. Service brakes shall be of the hydraulic, internal expanding type, or the disc type hydraulically actuated. The brake drum or housing shall be bolted, riveted, or cast integrally on each drive wheel or the brake shall be rigidly mounted on the pinion

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drive shaft adjacent to each wheel. Brakes shall be applied by depressing a pedal with the right foot. The brake pedal shall have a replaceable rubber cap. When actuated at all speeds, the brake shall provide smooth braking, free of brake chatter and free of feed-back amounting to more than 1/4 inch total movement measured at the foot pad. Either manual hydraulic or power brakes shall be provided to meet the performance requirements of 3.6.15 and 3.6.20. The hydraulic master cylinder shall be readily accessible for checking and refilling. When hydraulic master cylinder is located underneath the floor plate, an access hole not less than 2-1/2 inches in diameter shall be located above the master cylinder. The access hole shall be provided with a cover which can be easily removed. Brakes shall be designed to insure self-elimination of foreign material from the braking surfaces under normal maintenance and operating conditions and shall be adequately protected to insure satisfactory operation. Wheel cylinders shall be located so that hydraulic lines may be bled of air without the necessity of removing the wheels or disconnecting hydraulic lines. Brake lining, when used, shall be of the woven or molded type and shall be cement-bonded or riveted to the brake shoe. Service brake adjustment shall provide for at least 90 percent contact of brake lining with the drum throughout the service life of the brake lining. Brake adjustment mechanism shall be readily accessible and shall not require special tools. Self-adjusting brakes are acceptable. The master cylinder and brake pedal linkages shall be rigidly supported.

3.3.4.2 Parking brake. The parking brake shall be an independent mechanical friction brake mounted on the drive or propeller shaft rotating with the wheels, or a separate linkage actuating the service brake. It shall be hand-operated and equipped with a locking device. The manual control shall be located within easy reach of the operator and in a position permitting easy and safe movement on and off the vehicle from either side. The parking brake shall be capable of retaining the truck with rated load on a 15 percent grade, in both forward and reverse direction. If hand lever is of the ratchet type, the ratchet and pawl shall be of heat treated steel. Adjustment of parking brake shoes or bands shall be accomplished without the removal of any major assembly other than tires and wheels. This adjustment shall provide positive means to prevent loss of adjustment through inadvertent disengagement of adjusting device.

3.3.5 Body. The truck body (other than counterweight) shall be constructed of heavy steel sheet or plate equal to that used in the manufacturer's normal commercial product. Trucks having a horizontal floor plate in excess of 26 inches above ground level shall be provided with an intermediate step approximately midway between the ground and the floor plate. The step shall be large enough to accommodate a large foot (size 11) with an overshoe and shall not extend beyond the truck profile.

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3.3.5.1 Seat. A cushion seat and a cushion backrest, adjustable forward and reverse, shall provide comfortable seating for the operator when operating the truck. The seat shall have a minimum adjustment of 4 inches which shall be adjustable by the operator from the operating position without the use of tools. The top of the seat (not seat backrest) shall be not more than the specified dimension above the ground level. Effective dimension measurement point on seat to be the lowest point on the seat cushion in the depressed condition with a 150-pound operator seated in driving position.

3.3.5.2 Cowl or hood. The engine shall be completely enclosed with a hood cover and side plates. Either the hood cover or side plates shall be quick-opening. Hood cover, if lift-up type, shall be held in the open position in a safe manner by counterweight or spring and linkage mechanism or latch, which requires no brace or support and cannot be inadvertently released while servicing the vehicle. Hood cover shall be capable of being easily opened or removed by one man. Hand tools shall not be required for release of hood or side plates. All engine components (accessories) shall be completely accessible, without requiring the removal of any additional truck structure, when hood cover and side plates are opened. The hood shall be of such a design that when closed it shall prevent the entry of water on the engine electrical system when tested in accordance with MIL-STD-268.

3.3.5.3 Instrument panel. An instrument panel shall be installed convenient to, and plainly visible to the operator. The instrument panel shall be installed on an inclined or vertical plane to facilitate drainage of water. On this panel shall be mounted the ammeter, oil pressure gauge, coolant temperature gauge, fuel gauge, hour meter, ignition switch, light switch, and starter button or switch, and any other instruments, when used.

3.3.5.4 Wheel guards. If drive or steer wheels, when in a straight ahead position, protrude beyond the body of the trucks in excess of 1-1/2 inches, suitable wheel guards shall be provided to prevent foreign objects being discharged in the direction of the operator, endangering personnel, or otherwise creating a hazard.

3.3.5.5 Towing hook. A ring or pin type towing hook shall be provided in the center rear of the truck. It shall not protrude beyond the path followed by the rear end of the truck when making a turn of minimum turning radius. The towing hook shall be capable of withstanding, without physical failure, a towing force induced in accordance with Standard MIL-STD-268.

3.3.5.6 Lifting attachments. The truck shall be provided with four lifting eyes (2 forward and 2 rearward) to enable the truck to be lifted in its normal position by a cradle sling with cables vertical. No part

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of the truck shall be dislodged when truck is lifted clear off the floor by the cables. Lifting eyes shall be capable of withstanding tests in accordance with Standard MIL-STD-268. Inside diameter of each lifting eye shall be not less than 2 inches. The notation "LIFT HERE" shall be stenciled in close proximity to each lifting eye. Each lifting eye shall be capable of supporting 400 percent of the load normally applied to that lifting eye by a vertical cable with the truck in normal position.

3.3.5.7 Battery. Trucks shall be furnished with a 12-volt battery conforming to Specification W-B-131. Wood separators are not acceptable. The battery shall meet the starting requirements of 3.2.2.4. The battery shall be charged and dry. One filling of electrolyte of the required specific gravity shall be shipped with the truck in a separate container.

3.3.5.8 Battery mounting. Battery mounting shall be under the engine hood or in an accessible compartment for easy battery removal. Battery shall be mounted in such a manner as not to interfere with access to engine components (accessories), or provision shall be made to swing battery out of the way. Battery mounting shall provide for complete support over the entire base of the battery and shall be in such a position that the level of the electrolyte is directly visible without removing the battery from its mounting bracket or requiring the use of tools. Battery restraining clamps shall not be designed for or dependent upon special battery shapes and shall be provided to hold the battery in a fixed position. If mounted outside, the battery shall be housed in an individual acid resistant finished metal box, incorporating provision for drainage. The box shall be fitted with a quick release cover to provide for inspection and servicing. Cover and positioning shall be protected against short circuiting. The ungrounded cable shall be protected by rubber grommets or insulated passages at entry to battery box.

3.3.5.9 Battery cable terminals. Battery cables shall be in accordance with standard commercial practices. Clamp-type, secured with a bolt and nut, non-corrosive, terminal shall be provided on the ends of cables connected to the battery to facilitate disconnection and battery removal.

3.4 Controls and instrumentation. Where applicable, the provisions of Underwriters' Laboratories, Inc., "Standard for Safety (UL 558) shall apply. Electric controls and instrumentation shall be constructed to be moisture and weather resistant to prevent the entry of moisture when operated or stored out of doors under all weather conditions.

3.4.1 Controls. All switch and lever type controls shall be provided with position markings (decals are not acceptable) either at the switch, or lever, or in the case of the directional speed controls, on a diagram visible to the operator. The distance from the near edge of one knob to the near edge of an adjacent knob or other surface shall be a minimum of

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1-1/2 inches. All controls shall be easy to operate and positive in action.

3.4.1.1 Light switch. An "on-off" light switch shall be located on the instrument panel.

3.4.1.2 Ignition switch. The ignition switch shall be a keyless type. It shall have not less than 2 positions, an "on" and an "off". A second "on" position, for operation of electrical accessories, is permitted.

3.4.1.3 Starting motor switch. The starting motor shall be energized through a solenoid plunger type switch connecting the starting motor to the battery. Solenoid shall be operated by depressing a button installed on the instrument panel or turning a spring-loaded ignition switch to a "start" position. If a push button is used, the starter shall be inoperable when the ignition switch is in the "off" position.

3.4.1.4 Braking and inching pedal(s). Two pedals shall be provided for inching and braking control. The right pedal located for convenient right foot operation may provide braking control only or may be a combination inching-braking pedal. The left pedal located for convenient left foot operation shall be a combination inching-braking pedal. Each pedal shall be a minimum of 3 inches wide within a clear space of at least 6 inches wide. Grease fittings shall be provided for lubricating moving parts of all pedals and shafts. In lieu of two pedals for combination inching-braking, one combination inching-braking pedal not less than 8 inches wide may be provided and located for convenient right and left foot operation.

3.4.1.5 Accelerator control. The accelerator control shall be installed for comfortable right-foot operation and shall be located to the right of, and shall be free of interference from, the brake pedal by a distance of not less than 1-1/2 inches.

3.4.1.6 Directional speed controls. Selective forward and reverse (and speed range when used) speed controls shall be provided and shall be located for convenient right hand operation. Movement of the control lever in the forward or upward direction shall cause forward motion of the truck.

3.4.1.7 Lift and tilt controls. Lift and tilt controls shall be self-centering lever(s), conveniently located for right-hand operation. The tilt control shall be at the right of the lift control when 2 levers are employed. Lift and tilt controls may be combined in a single control lever. Rearward or upward motion of lift and tilt control lever(s) shall raise or tilt load rearward.

3.4.2 Instruments. All gauges and meters shall be products of manufacturers regularly engaged in producing these types of instruments. They shall be flush-mounted on the instrument panel.

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3.4.2.1 Ammeter. The truck shall be equipped with a remote indicating ammeter of sufficient capacity to indicate variations in current supply or demand experienced by the truck, exclusive of starter requirements.

3.4.2.2 Oil pressure gauge. A remote indicating electrical gauge shall be provided to indicate engine oil pressure.

3.4.2.3 Temperature gauge. A remote indicating electrical coolant temperature gauge shall be provided.

3.4.2.4 Hour meter. An electric operational hour meter, which registers the number of engine operating hours only, shall be provided. Hour meter shall be in accordance with Specification MIL-M-3971, type I.

3.4.2.5 Fuel gauge. An electric, thermally stabilized, remote indicating fuel gauge shall be provided.

3.5 Accessories. When applicable, the provisions of Underwriters' Laboratories, Inc., "Standard for Safety" (UL 558) shall apply. All accessories listed below shall be constructed to be moisture and weather resistant to prevent entry of moisture when operated or stored out of doors under all weather conditions.

3.5.1 Floodlight. Each truck shall be equipped with one sealed beam floodlight of not less than 25 watts, mounted on the left side of the mast. Directional focus of this light shall be vertically and horizontally hand adjustable without the use of hand tools. Adequate protection against damage by position or guard shall be provided. When specified (see 6.2), trucks shall be equipped with twin lights of not less than 25 watts, both front and rear, to provide adequate vision for all storage and loading operations. At least one rear and one front light shall be hand adjustable.

3.5.2 Stoplight. One automotive, red, reflector type, combination stoplight and tail lamp, of not less than 15 candle power and 3 candle power respectively, shall be recessed or mounted on the rear end within the plan outline of the truck. The stoplight shall not operate when ignition switch is in the "off" position. The light shall be protected against damage when mounted on the rear of the vehicle.

3.5.3 Side shift attachment. When specifled (see 6.2), trucks shall be equipped with a side shift attachment. Side shift mechanism shall be hydraulically-operated to move forks a minimum of 4 inches either side of center with rated capacity. The attachment shall be substantially constructed of welded steel. Visibility through the rack shall be maximum practicable. Fork hanger shall permit the addition of one extra fork at the fork carriage center line to provide 3 forks for bomb handling. Controls for the side shift mechanism shall be located for convenient

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right-hand operation and to the right of the tilt control. Fork, width, length and thickness shall be as specified. Forward, right or upward movement of the control shall cause the forks to move to the right.

3.5.4 Closed cab. Removable rigid closed cabs shall be provided when selected as an optional feature. The cab shall include windshield with shatter-proof glazing, two power-operated windshield wipers, complete with booster, and 2 side windows that can be opened. Cab glazing shall provide maximum visibility. A minimum clearance of 39 inches from top of seat to underside of top of cab shall be provided. Effective dimension measurement point in seat to be the lowest point on the seat cushion surface in the depressed condition with a 150 pound operator seated in driving attitude. Overhead guards need not be provided if cab top is reinforced to the same strength required for the overhead guard when tested in accordance with MIL-STD-268.

3.5.4.1 Cab heater. Cab heaters shall be of an automotive type space heater with windshield defrosting attachments when selected as an optional feature.

3.5.5 Load leveling device. When selected as optional equipment, the side shift attachment specified in 3.3.5.7 shall include a load leveling device to compensate for irregular ground surface conditions that may cause the vehicle to assume a laterally uneven position. The load leveling mechanism shall be hydraulically controlled and shall provide a minimum lateral tilt of the load carrying surfaces with and without rated capacity, 6 degrees each side of center. Controls for the load-leveling mechanism shall be located for convenient right-hand operation and to the right of the tilt control. Forward or upward movement of the controls shall cause load to rotate clockwise.

3.5.6 Horn. Truck shall be equipped with an electric horn with push button mounted in the center of the steering wheel. Horn button assembly and electrical wiring for the horn shall be constructed to be moisture and weather resistant to prevent entry of moisture when operated or stored outdoors under all weather conditions.

3.5.7 Driver's overhead guard. The overhead safety guard shall be fabricated from steel bars, tubing, or plate for protection of the operator. Where the upright in the collapsed position is lower than the overhead guard height, the guard shall be easily removable with hand tools. A solid plate is not acceptable. Distance between bars, tubing or plate, measured in both width and length, shall be no greater than 6 inches. Clear space shall be uniformly distributed and not less than 50 percent of total overhead area. Provisions shall be made so that failure of tilt cylinder assembly shall not cause overhead guard to injure the operator. A minimum of 39 inches from top of seat to underside of guard shall be provided at all tilt positions of the mast. Effective dimension measurement point on seat to be the lowest point on the seat cushion surface in the depressed condition with a 150-pound operator

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seated in the driving attitude. Guard height with mast tilted rearward, measured from ground level to top side of guard shall not exceed specified height. Guard design shall not interfere with operation of the truck, nor with normal movements of the operator when entering, leaving, or operating the truck. Overhead guard shall remain firmly secured to its support brackets under normal and test course operation.

3.5.8 Fire extinguisher. When specified (see 6.2), each truck shall be equipped with a fire extinguisher of the type and content prescribed. Mounting shall be on a positive locking, quick release bracket, in an accessible position, with the extinguisher nozzle protected against damage.

3.6 Performance.

3.6.1 Lifting speed. Speed of lift with rated capacity load shall be not less than 40 feet per minute over the entire distance from ground level to maximum fork height, measured in accordance with Standard MIL-STD-268.

3.6.2 Lowering speed. Speed of lowering of unloaded forks shall be not less than 27 feet per minute over the entire distance from maximum fork height to ground level, measured in accordance with Standard MIL-STD-268. The hydraulic control system shall be damped, metered, or easily manually regulated to reduce shock and prevent truck from overturning with rated load is lowered at maximum speed and then stopped quickly.

3.6.3 Right angle turn. Trucks when carrying rated capacity load, shall be capable of backing through a turn in either direction and returning to the same or a parallel position within dimension "A" as measured in Standard MIL-STD-268.

3.6.4 Steering. The steer wheels shall be capable of being turned from extreme right to extreme left or vice versa in not more than 6 complete turns of the hand steering wheel. When equipped with manual steer only the force required to operate within the first half of the inner steer angle (from a straight-ahead position) shall not exceed 40 pounds, nor shall it exceed 80 pounds at the extreme ends of the steer in either direction when the unloaded truck is stationary on a smooth, dry concrete floor as measured in Standard MIL-STD-268. When power assist is supplied, the maximum steering effort through the entire turn shall not be less than 2 pounds nor more than 10 pounds when measured under the same conditions. Power steering shall be effective at engine idling speed. Other maneuverability features shall meet test course requirements of Standard MIL-STD-268. Wear of the steering mechanism shall be measured in accordance with Standard MIL-STD-268. Free play shall not exceed 3 inches for manual steering and powered steering with engine at idle speed. The steering gear assembly shall be capable of withstanding the specified unbalanced tangential force applied at the rim of the hand steering

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wheel without permanently distorting, fracturing, or deforming any part of the steering mechanism as tested in Standard MIL-STD-268.

3.6.5 Speed. While carrying the rated capacity load on a level surface, trucks shall be capable of attaining specified speeds in high gear in the forward and reverse direction. Trucks shall be governed so that top speeds are within ranges specified. Speeds shall be measured in accordance with Standard MIL-STD-268.

3.6.6 Slope ascension, first speed, forward direction. Trucks shall be capable of ascending a 15 percent grade on a dry concrete surface with and without rated capacity load. Trucks shall be able to accelerate from a dead stop on this slope, when carrying rated load, as measured in Standard MIL-STD-268.

3.6.7 Upright tilt. When unloaded, truck masts shall have the forward tilt and backward tilt specified. Tilt shall be measured from a true vertical, in accordance with Standard MIL-STD-268.

3.6.8 Upright height. Collapsed mast height of unloaded truck measured from the ground to the top of the uprights in a vertical position shall not exceed those specified.

3.6.9 Fork height. The fork height, with truck bearing rated capacity load, and the uprights fully extended and in vertical position, measured from the ground to the load carrying surface of the fork, shall not be less than that specified. The position of measurement on the forks shall be at rated load center.

3.6.10 Free lift height. Height of forks without increase in the specified collapsed mast height shall be not less than that specified. Free lift height shall be measured in accordance with Standard MIL-STD-268.

3.6.11 Underclearance. Truck, with rated capacity load, shall have sufficient underclearance to permit operation from one horizontal plane to another, up or down a 15 percent slope and the obstacle course described in Standard MIL-STD-268. Minimum underclearance beneath the mast assembly, when in true vertical position shall be not less than that specified when the truck is loaded.

3.6.12 Lift assembly. The lift assembly shall show no appreciable signs of wear after completing the endurance requirements of the test course under the provisions of Standard MIL-STD-268. "Appreciable wear" is defined as any apparent inability to lift and hold rated load for a period of 2 minutes with not more than 1 inch drift. Packing nut (if used) adjustment will not be permitted during test prescribed in Standard MIL-STD-268.

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3.16.3 Forks, upright, and carriage assembly. The assembly and related structures shall show no permanent deformation or failure following test under 300 percent overload when tested in accordance with MIL-STD-268.

3.6.14 Hydraulic system. The hydraulic system shall show no leaks or breakage when the truck is subjected to the test prescribed in 3.6.13 with the pressure relief valve inoperative. This requirement does not include the pump. The pump relief valve shall open when pressure equals no more than 150 percent of the maximum pressure needed to lift rated load to maximum fork height.

3.6.15 Service brakes. A brake pedal pressure of not more than 125 pounds and not less than 50 pounds shall develop a drawbar pull equal to 30 percent to the gross weight of the truck (with rated load) while the truck is being towed over a concrete surface having a coefficient of friction not less than 0.50. The brake system shall be capable of withstanding brake pedal pressure of 250 pounds without failure of any component. These requirements shall be measured in accordance with Standard MIL-STD-268.

3.6.16 Longitudinal stability.

3.6.16.1 Stacking. Using the procedure outlined in Standard MIL-STD-268, the truck carrying rated capacity at the maximum fork height shall not overturn when the platform upon which the truck is placed is tilted to the slope on the applicable specification sheet.

3.6.16.2 Traveling. Using the procedure outline in Standard MIL-STD-268, the truck carrying the rated capacity at 12 inches above the floor shall not overturn when the platform upon which the truck is standing is tilted to the slope on the applicable specification sheet.

3.6.17 Lateral stability.

3.6.17.1 Stacking. Using the procedure outlined in Standard MIL-STD-268, the truck carrying the rated capacity at the maximum fork height shall not overturn when the platform upon which the truck is standing is tilted to the slope on the applicable specification sheet.

3.6.17.2 Traveling. Using the procedure outlined in Standard MIL-STD-268, the truck without load shall not overturn when the platform upon which the truck is standing is tilted to the slope on the applicable specification sheet.

3.6.18 Driver's overhead guard. The overhead safety guard shall be capable of withstanding the impact of a 100 pound, solid hardwood cube

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(or equivalent) dropped a distance of 5 feet 10 times, without fracture or without permanent deflection exceeding 3/4 of an inch. It shall be of sufficient strength to support an uniformly distributed static test load equal to the applicable loads. When a high strength overhead safety guard is specified, the guard shall be capable of withstanding an 8 foot drop of 500 pounds high density cube 10 times without permanent deformation exceeding 3 inches.

3.6.19 Acceleration. From a standing start, the truck, when carrying rated load, shall have an acceleration in the forward direction such as to negotiate the specified distance in not more than the designated time. Compliance shall be measured in accordance with Standard MIL-STD-268. There shall be not more than one second delay in the movement of the truck when tested in accordance with Standard MIL-STD-268.

3.6.20 Stopping distance. Trucks, with rated load, shall be capable of stopping in reverse from maximum speed of the vehicle in an average distance not greater than the appropriate distance given in American Standard Safety Code for Powered Industrial Trucks (ASA B56.1), when a brake pressure as specified in paragraph 3.6.15 is applied in accordance with Standard MIL-STD-268.

3.6.21 Parking brake. The parking brake shall be capable of retaining truck with rated load on the grade specified in 3.3.4.2 in both forward and reverse direction when tested in accordance with Standard MIL-STD-268.

3.7 Workmanship. Finished items shall be clean, well made, and free from any defects which may affect appearance or serviceability. All unnecessary pans, pockets, reservoirs, etc., where grease, oil and dirt might collect, shall be eliminated.

3.7.1 Rivet connections. Rivet holes, in the sizes recommended by standard practice, shall be accurately punched or drilled, and shall have burrs removed. Rivets shall be driven with power tools. Rivet heads shall be full, neatly made, concentric with rivet holes, in full contact with the surface of the member. Excessive upsetting of rivets to fill holes will not be acceptable.

3.7.2 Welds. Welded joints shall be sound, smooth, free from pits, holes, fissures, rough projecting edges, and slag; and shall be thoroughly fused to the base metal.

3.7.3 Bolt connections. Bolt holes shall be accurately punched, drilled, reamed or cast and adjacent edges and surfaces shall have no burrs or fins or irregularities to interfere with proper seating of bolt head or nuts. Internal parts of the engines, transmissions and drive trains, and parts subject to excessive or critical vibrations assembled with bolts and nuts, cap screws or studs, or set screws, shall

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have such fasteners secured with a positive locking device. This device may be, but shall not be limited to, lock-washers, cotter pins, safety wire, roll pins, or anti-vibration self-locking nuts. All nuts shall have full thread engagement when properly tightened.

3.8 Treatment and painting. All exterior surfaces of the truck shall be thoroughly cleaned and shall be dry and free from mill scale, oil, grease, dirt and rust and shall be painted as soon as practicable after cleaning with not less than one coat of metallic base primer and two coats of snythetic enamel. Primer and finish coats shall be applied in accordance with normal commercial practices to assure complete coverage and durability of the finish. The finish color shall be 13538 in accordance with Federal Standard No. 595. When a preproduction model is required for test and evaluation in accordance with Standard MIL-STD-268, it shall be painted with one primer coat only. The finish coat, when dry, shall be a smooth even surface, free from runs, sags, peels, chips, blisters, areas of thin film or no film. Surfaces of components and assemblies of the truck, not normally painted with a finish coat shall be cleaned and prepared in accordance with good commercial practice. Lift chains, sprockets, rollers, plastics, fabric hoses (except hydraulic hoses) and drive belts shall not be painted except that slight overspray will be accepted provided there is no interference with the proper functioning of the component. Paint must be kept off working surfaces where interference with working parts would result. Engine accessories, i.e., generator, starter, radiator, carburetor, engine ignition system components and instruments, plated or painted by the manufacturer of such items shall not be painted by the contractor and shall be adequately protected or masked from overspray.

3.8.1 Walkway coating. Floor plates and step surfaces shall be coated or matted with type II or III, color 604, black, of Specification MIL-W-5044.

3.9 Radio interference suppression.

3.9.1 Army and Navy. Equipment shall be designed and equipped to the administrative requirements in accordance with Specification MIL-E-55301, except that when specified, equipment intended for use aboard ship, advance bases, or in the vicinity of Naval electronic installations, radio interference suppression shall be in accordance with Specification MIL-I-16165.

3.9.2 Air Force. Radio interference suppression of equipment procured for Air Force use shall be in accordance with MIL-STD-826.

3.10 Nameplate. The manufacturer shall provide and install a corrosion-resistant metal nameplate, according to Standard MIL-STD-130, (for Department of the Navy (less U. S. Marine Corps), stock numbers are not

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required), in a visible and safe location on the instrument panel or fire wall of the truck, with the following information indented or embossed on the plate:

TRUCKS, LIFT, FORK GASOLINE

CAPACITY.....pounds at ___ inch load center.
NAME OF MANUFACTURER.....
FSN NUMBER.....
SERIAL NUMBER.....
MODEL NUMBER.....
REGISTRATION NUMBER.....
U. S. ARMY MODEL NUMBER.....(when applicable)
DELIVERY DATE.....(month and year)
WARRANTY EXPIRATION DATE.....(month and year)
CONTRACT OR ORDER NUMBER.....
SERVICE WEIGHT.....pounds
WHEEL LOADING (no load on forks)
  DRIVE WHEELS (each).....pounds
  STEER WHEELS (each).....pounds
*WHEEL LOADING (rated load on forks)
  DRIVE WHEELS (each).....pounds
  STEER WHEELS (each).....pounds
CENTER OF GRAVITY (no load on forks)
  HORIZONTAL.....inches from axle of drive wheels.
  VERTICAL.....inches above axle of drive wheels
IDENTIFICATION OF MAINTENANCE MANUAL.....
  (truck serial number or other)

U. S.
NOTE: *Mast vertical and load in carrying position.

3.11 Registration data. The manufacturer shall execute registration data cards, as required by the contracting officer. Unless otherwise specified, the assigned registration number for each truck shall be painted by the contractor, in black enamel, in block numbers 3 inches high on each side of the upright mast.

3.12 Technical manuals. Unless otherwise specified, one copy of the technical manual conforming to Specification MIL-M-21861 shall be furnished by the manufacturer with each truck at time of shipment. The manual shall be placed in a protected place in the equipment. Each manual shall be preserved and packaged by Method IC-1 of Specification MIL-P-116 and the bags heat sealed. Additional copies of the manual shall be furnished as specified.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the end item manufacturer may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample inspection. Inspection shall be made of a completely assembled unit in accordance with the detailed requirements of Standard MIL-STD-268 as specified in 4.3.3.2.1.

4.3 Sampling for inspection and acceptance. When sampling for inspection and acceptance is performed, it shall be performed in accordance with the provisions set forth in Standard MIL-STD-105, except where otherwise indicated.

4.3.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification.

4.3.2 Process examination. Intermediate assemblies which cannot be checked or measured after final assembly shall be examined at various stages of manufacture for compliance with this specification. Whenever a deviation is noted, correction shall be made. Failure to make immediate correction may cause rejection of affected end items.

4.3.3 End item inspection.

4.3.3.1 Examination. Each fully assembly unit shall be examined for completeness of assembly, presence and adequacy of components, repair parts, operator's tools (when required), technical manuals and accessories, and quality of workmanship to assure that the unit is manufactured in accordance with specified requirements. Examination shall be made for items on the Pretest Inspection Check-List of Standard MIL-STD-268. Whenever a deviation to specified requirements is noted, corrective action shall be made. Failure to make corrections shall cause rejection of the unit.

4.3.3.2 Testing of end item.

4.3.3.2.1 Preproduction testing. The preproduction sample truck shall be tested for all provisions of this specification using MIL-STD-268 and

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4.4. These tests shall be witnessed by the contracting officer or his authorized representative or appointed test officer. Failure of the preproduction sample to pass these tests shall be cause for rejection of the preproduction sample. If a change is made in design, material or construction, subsequent to the approval of the preproduction sample, retest in accordance with 4.4 and MIL-STD-268 shall be required.

4.3.3.2.2 Testing of production items. Each production unit shall be tested to determine compliance with paragraphs 3.6.1, 3.6.2, 3.6.5, 3.6.6, 3.6.11, 3.6.19, 3.6.20 and 3.6.21 using MIL-STD-268. During these tests, all gauges, instruments, lights, horns, inching mechanisms and other operating components shall be tested to determine that they perform the function for which they are intended. Any failure to meet test requirements shall be cause for rejection of the truck under test. In addition to the above tests, the supplier shall furnish a certificate of compliance for starting system, transmission and drive assembly, belts and engines, and attachments (when required), certifying that the component complies in all particulars with the requirements of this specification and subsidiary specifications.

4.3.3.2.3 In addition to tests specified each production unit in paragraph 4.3.3.2.2 above one (1) unit shall be tested in accordance with all provisions of this specification except that testing for conformance with paragraph 3.3.1, 3.3.2.1, 3.1.3.7 and 3.2.3.3 shall be done only when specified. The unit to be tested shall be designated by an authorized representative of the contracting officer. Failure to meet any of the specified requirements shall be cause for testing of all trucks on hand for the characteristic(s) in question.

4.3.3.3 Code and standards compliance. The supplier shall furnish the Government evidence of compliance with paragraph 2.2.

4.3.4 Examination of preparation for delivery. An examination will be made to determine that requirements for preservation, packaging, packing and marking as requirement by Standard MIL-STD-162 are complied with.

4.4 Tests.

4.4.1 Radio interference suppression. Test to determine compliance with 3.9 shall be performed in accordance with the requirements of the applicable specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be level A, B or C, in accordance with MIL-STD-162 as specified (see 6.2).

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5.2 Marking. When applicable, in addition to any special marking specified in the invitation for bid, marking shall be in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. Trucks described herein are intended for stacking, unstacking, and moving cargo in and around warehouses, loading platforms, and docks within the military supply system; also for moving cargo in and out of highway trailers and railroad cars. Trucks are intended for operation over paved, semiprepared and other hard surfaces for short distances.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this document and the applicable specification sheet.
(b) Accessories and optional equipment selected from list on specification sheet.
(c) When a preproduction model is not required (see 3.1.1).
(d) When a commercial engine shall be supplied (see 3.2.1).
(e) When production control tests, production, and inspection are to be other than commercial practice (3.2.1.1).
(f) When transmission and drive assembly test is required in lieu of certificate of compliance (see 3.2.3).
(g) When mechanical transmission is required (see 3.2.3.1).
(h) When frame racking test is not required (see 3.2.1).
(i) Radio interference suppression data when required (see 3.9).
(j) Type of registration data required (see 3.11).
(k) Additional distribution of technical manual when required (see 3.12).
(l) Level of preservation, packaging and packing required (see 5.1).
(m) Special marking requirements (see 5.2).

Custodians:

Army - GL
Navy - SA
Air Force - 84

Preparing Activity:

Navy - SA

Project No. 3920-0102

Interested Activities

Users and Review - Army - SM - GL
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